

Semanur Avsar

☎ +41-76-244-91-15 ✉ semanur.avsar@epfl.ch 🏠 Lausanne, Switzerland

🌐 linkedin.com/in/semanuravsar 🐙 github.com/semanuravsar

Education

École Polytechnique Fédérale de Lausanne (EPFL) <i>Master of Science in Electrical and Electronics Engineering; GPA: 5.60/6.00</i>	Lausanne, Vaud, Switzerland 2024 – Ongoing
Middle East Technical University (METU) <i>Bachelor of Science in Electrical and Electronics Engineering; GPA: 3.93/4.00</i>	Ankara, Turkey 2019 – 2024
Universität Paderborn <i>Erasmus exchange student at the Department of Electrical Engineering</i>	Paderborn, Germany 09/2022 – 03/2023

Research Experience

Graduate Researcher Signal Processing Laboratory (LTS4) • Integrated phase-based functional connectivity (Phase-Locking Value, PLV) into a graph dictionary learning framework for EEG motor imagery decoding. • Implemented and compared multiple PLV-based model variants, analyzing the effects of distance transformations, window length, frequency-band selection, scaling, and coefficient thresholding on learned graph atoms. • Designed and ran extensive experiments on the EEGBCI dataset using group-aware cross-validation and multi-classifier evaluation.	Lausanne, Switzerland 09/2025 – 01/2026
Graduate Researcher Telecommunications Circuits Laboratory (TCL) • Developed structured sparse parity-check matrices to improve belief propagation (BP) decoding of short LDPC codes. • Developed an NSGA-II based multi-objective optimization framework to tune matrix parameters using block-wise genetic operations. • Explored multiple objective formulations, including rank, 4-cycle count, and BER performance, to guide multi-objective search.	Lausanne, Switzerland 02/2025 – 06/2025
Summer Intern Adaptive Systems Laboratory (ASL) • Selected for the E3: EPFL Excellence in Engineering summer research fellowship. • Contributed to the development of Python codes for the simulations and visual data of the book Inference and Learning from Data Volume 1: Foundations. • Created LaTeX-formatted Jupyter notebooks to support course-based learning from the book.	Lausanne, Switzerland 07/2023 – 09/2023
Undergraduate Researcher Communication Networks Research Group (CNG) • Implemented and simulated AoI and QAOI minimization algorithms in MATLAB under i.i.d. and Markovian transmission delays. • Evaluated algorithm performance across various delay distributions (exponential, lognormal, Pareto). • Modeled correlated random variables and Markov chains to simulate realistic delay dynamics in queue-based systems. • Contributed to a conference paper promoting QAOI as a metric for real-time system design.	Ankara, Turkey 07/2021 – 07/2022

Work Experience

Research Intern NAGRA Kudelski Group • Developing an adaptive anti-jamming system on SDR platforms (GNU Radio, Python) that monitors the radio spectrum, identifies jamming patterns, and autonomously switches frequencies to maintain reliable communication. • Implementing a tabular Q-learning agent and a Deep Q-Network (DQN) agent with multiple state encodings derived from the raw spectrum to learn channel-selection policies against different types of jammers.	Cheseaux-sur-Lausanne, Switzerland 02/2025 – Ongoing
Part-Time Engineer Plan-S Satellite and Space Technologies • Worked on waveform design and testing in the Telecommunications Design Department. • Conducted channel simulations and waveform analysis using MATLAB, GNU Radio, and Python. • Developed complete GNU Radio blocks for LR-FHSS modulation using Python.	Ankara, Turkey 02/2024 – 07/2024
Summer Intern Aselsan Military Electronics Industries • Worked in the Waveform Design Department, designed and evaluated Lateration and Min-Max algorithms for ad-hoc network localization in MATLAB, under static and mobile node conditions. • Studied the impact of anchor placement, measurement count, and mobility; proposed enhancements like collinearity checks and iterative localization.	Ankara, Turkey 07/2022 – 08/2022

Publication

- M. E. Ildiz, S. Avşar, E. Uysal, "An Inequality for Query Age of Information and Age of Information," *30th Signal Processing and Communications Applications Conference (SIU)*, May 2022.

Selected Course Projects

Multilingual Web Document Genre Classification for LLM Data Curation [Report](#) Fall 2025

- Built a multilingual web document genre classifier (8 classes) for LLM pre-training data curation, systematically comparing traditional (TF-IDF + Logistic Regression, FastText) and transformer-based approaches (BERT, XLM-RoBERTa) under a unified evaluation pipeline.
- Designed a data cleaning pipeline using Jaccard similarity to detect and merge near-duplicate samples, reducing label noise across 14,619 text snippets and improving label reliability for the majority of samples.
- Evaluated monolingual vs. multilingual training across 7 languages, finding that naive machine-translated augmentation degraded performance, informing best practices for multilingual LLM corpora.

Graph-based EEG Seizure Detection [Report](#) Spring 2025

- Modeled EEG signals as dynamic graphs and implemented various graph neural network (GNN) architectures (GCN, ChebNet, GAT, GraphSAGE) for seizure classification.
- Designed diverse preprocessing pipelines (STFT, wavelet, bandpower) and graph construction strategies (mutual information, inverse distance) to extract informative signal and relational features.
- Systematically tested combinations of preprocessing methods, graph construction techniques, and GNN models to identify the best-performing setup for seizure detection.

Multi-Task Learning for Implicit Hate Speech Detection [Report](#) Spring 2025

- Proposed and implemented a multi-task learning framework to enhance implicit hate speech detection and improve out-of-domain generalization capabilities
- Developed a BERT-based model integrating auxiliary tasks such as sarcasm detection, stereotypical bias identification, and fine-grained subtype classification to enrich contextual understanding
- Designed and executed comprehensive experiments, including an ablation study, to assess the contribution of each auxiliary task to overall performance

OFDM system to transmit and receive data using a loudspeaker and microphone [Report](#) Fall 2024

- Developed and implemented an OFDM system from scratch in MATLAB for data transmission and reception via a loudspeaker and microphone.
- Integrated key OFDM components, including modulation, IFFT/FFT, cyclic prefix handling, and channel equalization.
- Enhanced the system with additional features, such as Hamming error correction and scrambling.

Ad Hoc Communication Network System for Disaster Relief [Report](#) Fall 2023 – Spring 2024

- Designed a mobile multi-robot communication system for disaster environments, implementing an infrared-based ad hoc network with CRC-checked packets, directional scanning, and a lightweight protocol for reliable inter-robot coordination.
- Developed the robot decision-making and navigation logic, including tile-based mapping, localized search, collision avoidance, and coordinated task updates between mobile and base units.
- Built and tested a working prototype as part of a six-member team, integrating communication and control layers to demonstrate dynamic multi-agent collaboration in a simulated search-and-rescue scenario.

Skills

- **Programming:** Python, C, Verilog HDL
- **ML & Deep Learning:** PyTorch, PyTorch Geometric, HuggingFace Transformers, scikit-learn, FastText, OpenCV, torchvision
- **Data & Visualization:** NumPy, Matplotlib, Jupyter
- **Signal Processing & Hardware:** MATLAB (incl. Simulink), GNU Radio, FPGA, LTspice
- **Languages:** Turkish (native), English (proficient), German (beginner), French (beginner)